EIACD Project
(Agentic AI for Deployment of ML Models)

Project Goal

The goal of this project is to develop a Agentic AI system for the deployment of ML models using KNIME.

Learning Goals

Consolidate, apply and combine concepts from the Artificial Intelligence and Data Science content of the course.

Recommendations

- Use a supervised learning model (either classification or regression). It is possible to use unsupervised models (either clustering and classification) but it may be harder to come up with tools.
- Use the exercise from the lab session on agentic AI as the basis for the development.
- Start by developing an application with a single tool, to make a prediction for new data stored in a table. Then, add a second tool to obtain the confidence of the prediction. Then, add the possibility for new data to be described in plain text.
- After the previous system is working, add 1 or 2 additional tools that you find interesting.
- Usage of LLMs to assist in the developmen of the project should be made carefully. For instance, the Claude model from IAEdu is very helpful but can also be misleading.

Groups

Groups should be constituted by 3 students, preferably from the same lab session. The students in each group should be indicated in moodle, in the activity provided for that purpose.

Evaluation

Each group will:

- make a demo of their solution during the last lab session of the semester.
- have 5min to make the presentation, followed by 5min of discussion.
- submit in moodle the application that was developed and a short powerpoint presentation (max. 3 slides + cover slide) explaining how to use it.